

Autonomous Racer

Category: Autonomous GrandPrix

Solution: Autonomous 1:10 Scale High-Speed Circuit Navigator

Developer: Respect Kagiso Morupane

Institution: Mopani TVET College (Mopani Campus)

Context and Problem Statement

Track & Autonomous Constraints

- ▶ **Autonomous Navigation:** High-speed trajectory calculation without human steering or remote control.
- ▶ **Real-Time Processing:** Low-latency wall detection under strict computational time limits.

Competition Rule Constraints

- ▶ **Weight Cap:** Maximum 5.0 kg total mass constraint.
- ▶ **Size Cap:** Must fit within a 50 cm x 50 cm footprint.
- ▶ **Budget Cap:** Total build expenditure under R 3,000.00.
- ▶ **Safety Mandate:** Hardware-level E-Stop disarm button and physical master power switch.

Solution Overview & Marketability

Technical Architecture

- ▶ **Microcontroller:** ESP32-WROOM-32U dual-core running C++ firmware.
- ▶ **Perception Array:** TF-Luna LiDAR (ToF range: 0.2m–8m) + Dual Sharp IR analog sensors.
- ▶ **Drive System:** High-speed 3650 3900KV brushless motor with a 60A ESC.
- ▶ **Steering System:** DS3218MG 20kg.cm metal-gear digital servo on Ackerman geometry.

Marketability & Buildability

- ▶ **Cost-Effective:** Complete system built under budget at R 2,990.00.
- ▶ **Modular Design:** Swappable PETG 3D-printed mounts for rapid hardware repairs.
- ▶ **Scalable Codebase:** FreeRTOS task isolation allows firmware porting to larger autonomous vehicles.

Mechanical & Electronic Hardware Systems

Mechanical & Chassis Design

- ▶ **Chassis:** 1:10 scale rear-wheel-drive (RWD) on-road touring frame.
- ▶ **Mass Distribution:** Low center-of-gravity with a 45/55 rear-biased weight balance for traction.
- ▶ **Custom Mounting:** PETG 3D-printed mounts elevating LiDAR to 100 mm to eliminate floor bounce.

Power & Electronic Integration

- ▶ **Dual-Rail Power:** 7.4V direct LiPo rail for ESC; regulated 5.0V Buck converter rail for ESP32 and sensors.
- ▶ **Master Safety:** Inline 20A heavy-duty isolation switch on the main battery lead.
- ▶ **Hardware E-Stop:** Red latching push button on GPIO 23 level-1 interrupt to kill motor signal instantly.

Financial Breakdown & Total Implementation Cost

Component Category	Key Items Selected	Cost (ZAR)
Chassis & Drivetrain	1:10 RWD Frame, 3650 Brushless Motor + 60A ESC	R 1,300.00
Sensors & Perception	TF-Luna Micro LiDAR + Dual Sharp GP2Y0A21YK0F IR Sensors	R 835.00
Power, Logic & Safety	ESP32-32U, 2S LiPo, Buck Converter, Toggle & E-Stop Switches	R 425.00
Steering & Mounting	DS3218MG 20kg Servo, PETG Prints, Fasteners & Wiring	R 430.00
FINAL TOTAL SPEND	FULLY COMPLIANT WITH COMPETITION BUDGET	R 2,990.00

Total budget allowed: R 3,000.00. Remaining buffer: R 10.00.

Software Architecture & Mentorship Validation

- ▶ **FreeRTOS Dual-Core Tasks:**

- ▶ **Core 0 (250 Hz Loop):** UART LiDAR packet parsing and ADC dual IR sensor reading.
- ▶ **Core 1 (100 Hz Loop):** Proportional-Integral-Derivative (PID) algorithm calculating steering angles and speed adjustments.

- ▶ **Fail-Safe Firmware Integration:** GPIO 23 ISR instantly forces ESC PWM to 1500 μ s (neutral disarm) on E-Stop activation.

- ▶ **Technical Excellence & Mentorship:**

- ▶ Verified participation in Facilitator Q&A (FQA) sessions.
- ▶ Complete FQA Logbook (FQA_Attendance_Log.md) submitted to claim 10 bonus points.

Lead Developer (Individual)

Respect Kagiso Morupane

Role: Systems Engineer, Firmware Developer & Fabricator

Institution: Mopani TVET College (Mopani Campus)

Deliverables Completed: Folder A (Code/Logic), Folder B (Cad/Wiring), Folder C (Business/Docs)

Repository: RoboRumble-2026-Autonomous-GrandPrix on GitHub

